Optimal Assembly Line

AlgoExpert · Very Hard · Searching

The problem

- Split step durations (in order) into k contiguous stations minimising the longest station time.
- Binary search on the answer with a greedy feasibility check.

Intuition

- Binary search the maximum station time; greedily check whether k stations suffice.

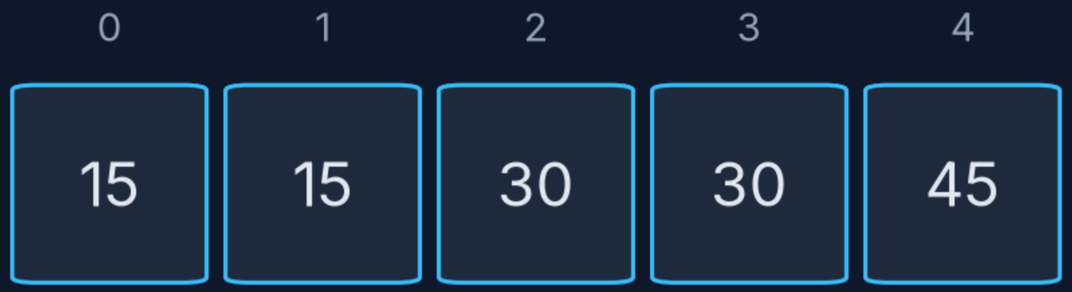
Brute force vs optimal

- Try all partitions → exponential
- DP → $O(k \cdot n^2)$
- Binary search on the answer + greedy check → $O(n \log \Sigma)$

Approach

- 1. Bounds are the largest step and the total
- 2. Greedily fill stations
- 3. Shrink the range when feasible

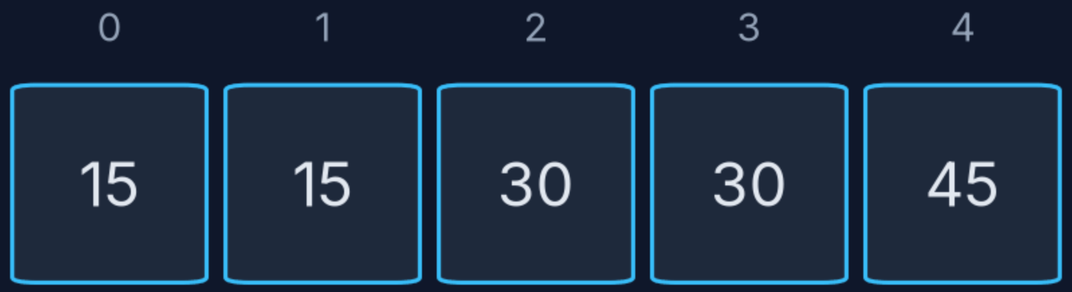
Worked example




```
max time 90: needs 2 stations
range [45, 135] → hi = mid
```

Try 90: 2 stations

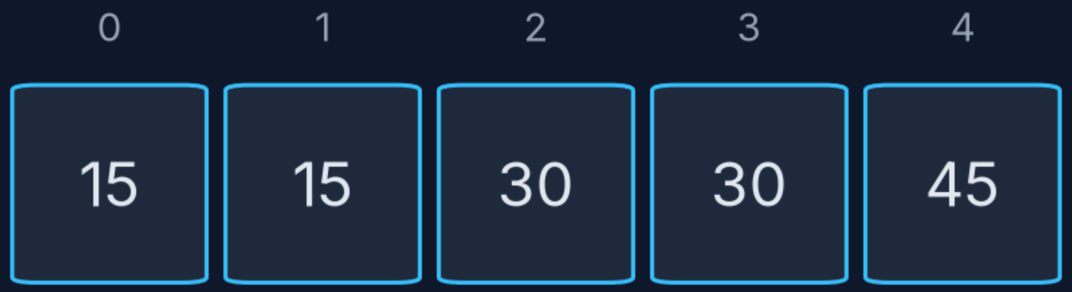
Worked example




```
max time 67: needs 3 stations
range [45, 90] → hi = mid
```

Try 67: 3 stations

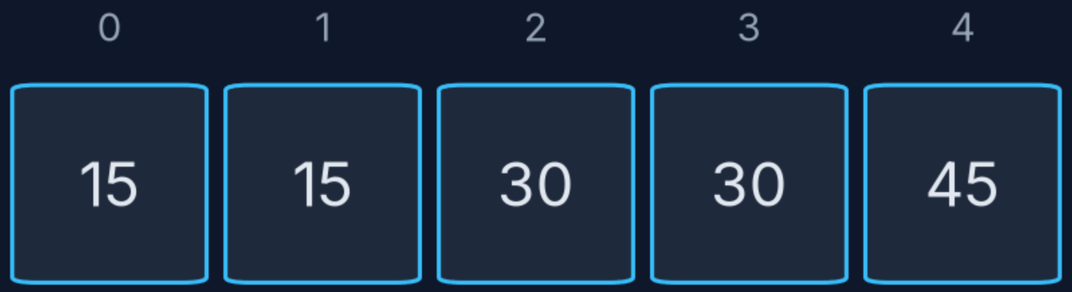
Worked example




```
max time 62: needs 3 stations
range [57, 67] → hi = mid
```

Try 62: 3 stations

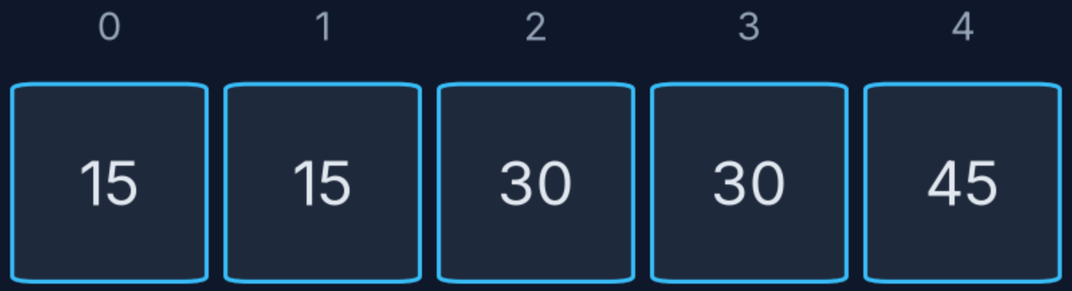
Worked example




```
max time 59: needs 4 stations
range [57, 62] → lo = mid + 1
```

Try 59: 4 stations

Worked example




```
max time 60: needs 3 stations
range [60, 61] → hi = mid
```

Try 60: 3 stations

Worked example

```
answer: 60
```

Answer: 60

Java solution

```java
public class OptimalAssemblyLine {
    public static int optimalAssemblyLine(int[] steps, int k) {
        int lo = 0, hi = 0;
        for (int s : steps) { lo = Math.max(lo, s); hi += s; }
        while (lo < hi) {
            int mid = (lo + hi) >>> 1;
            if (stations(steps, mid) <= k) hi = mid; else lo = mid + 1;
        }
        return lo;
    }

    private static int stations(int[] steps, int cap) {
        int count = 1, cur = 0;
        for (int s : steps) { if (cur + s > cap) { count++; cur = 0; } cur += s; }
        return count;
    }
}
```

Complexity

- Time: O(n log S)
- Space: O(1)

Edge cases & pitfalls

- Lo = max step
- Greedy check is optimal for a fixed cap

Interview tips & follow-ups

- Same as Split Array Largest Sum

Recap

- Binary search the maximum station time; greedily check whether k stations suffice.
- Complexity: O(n log S)